

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Summer Examination – 2025****Course:** B. Tech **Branch:** Civil Engineering **Semester :**VI**Subject Code & Name:** BTCVC601_Y18 & Design of Concrete Structures I**Max Marks:** 60**Date:** 30/06/2025**Duration:** 3 Hr.**Instructions to the Students:**

1. Each question carries 12 marks.
2. **Question No. 1 will be compulsory and include objective-type questions.**
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Minimum grade of concrete for PCC is ----- a. M-20 b. M-25 c. M-15 d. M-10	1	1
2	Minimum no. of bars to be used in circular column is ----- a. 2 b. 3 c. 4 d. 6	1	1
3	Minimum diameter of bar used in steel reinforcement is-----mm. a. 3 b. 5 c. 6 d. 8	2	1
4	Minimum diameter of bar to be used for column is -----mm a. 10 b. 12 c. 16 d. 20	2	1
5	(x_u/d) limiting value for Fe-415 is ----- a. 0.53 b. 0.48 c. 0.55 d. 0.46	3	1
6	Beam is design as doubly if M_u ----- $M_{u, \text{limiting}}$ a. equal to b. more than c. less than d. none of these	3	1
7	Single flight of a dog-legged stair is -----slab a. one way b. two way c. cantilever d. none of these	4	1
8	$M_{u, \text{limiting}}$ for Fe-415 is ----- $f_{ck} b d^2$ a. 0.133 b. 0.138 c. 0.148 d. _____	4	1
9	Vertical stirrups in beams resist ----- a. B.M b. SF c. Torsion d. All of these	5	1
10	LL for public building is -----kN/m ²	5	1

	a.1	b.2	c. 3	d. 46		
11	Position of main steel in a cantilever beam to resist BM is at ----				6	1
	a. bottom mid section	b. bottom support section	c. top mid section	d. top support section		
12	Load carrying capacity of a rectangular column is -----% less than that of circular column.				4	1
	a. 10	b.15	c.5	d. 25		
Q. 2	Solve the following.					12
A)	What do you mean by over- reinforced beam? Design by LSM, a simply supported beam, 3 m in span subjected to u.d.l of 20 kN/m over its entire span and a central point load of 15 kN. Use M-25 grade concrete, Fe-415 grade steel and 20 mm diameter bars. Adopt width of beam equal to 300 mm. Given $d'=25$ mm				1	6
B)	Explain role of longitudinal steel and lateral ties in columns. Design a short RCC column to carry an axial ultimate load of 1000 kN. Clear height of column is 3.1 m. Use M-20 grade concrete and Fe-415 grade steel. Draw a neat diagram of column (longitudinal section and cross-section) to show reinforcement details.				4	6
Q.3	Solve the following.					12
A)	Design by LSM a slab for a passage 3.1 m wide, supported over two opposite beams, each 230 mm wide. The LL & FF on the slab is 3kN/m ² and 0.5 kN/m ² respectively. Use M-20 grade concrete and Fe-415 grade steel. Draw neat sketches to show reinforcement details in cross-sections (short span and long span) and top view.				3	6
B)	Design a dog-legged stair for floor to floor height 3.2 m. Take LL and FF equal to 3 kN/m ² and 1 kN/m ² . Use M-20 grade concrete, Fe-415 grade steel and 20 mm diameter bars. Width of each flight is 1.0 m. Draw neat diagrams to show reinforcement details in longitudinal section and plan.				5	6
Q. 4	Solve Any Two of the following.					12
A)	Under what situations a doubly reinforced beam is adopted? Design by LSM ,a simply supported beam, 5.5 m in span subjected to u.d.l of 60 kN/m over its entire span. Use M-25 grade concrete, Fe-500 grade steel and 16 mm diameter bars. Adopt Size of beam equal to 300 mm x 500 mm (Overall). Given $d'=30$ mm for compression and tension reinforcements. Draw a neat diagram of beam (longitudinal section and cross-section) to show reinforcement details.					6

B)	Explain structural behavior of a one-way slab. Design by LSM, a slab for a room 3m x 4m (inner dimensions) supported over four beams, each 300 mm wide. The LL & FF on the slab is 2 kN/m ² and 0.5 kN/m ² respectively. Use M-20 grade concrete and Fe-415 grade steel. Assume slab is continuous over all edges. Draw neat sketches to show reinforcement details in cross-sections (short span and long span) and top view.	4	6
C)	A simply supported beam, 3 m in span subjected to u.d.l of 20 kN/m over its entire span. Use M-20 grade concrete, Fe-415 grade steel and 12 mm diameter bars. Adopt size of beam equal to 300 mm x 400 mm (overall). Given $d'=25$ mm. Beam is reinforced with 3 bars of 16 mm at support and mid -section. Design the beam for shear reinforcement. Draw neat diagrams to show reinforcement details in longitudinal section and cross-section. Adopt working stress method .	3	6
Q.5	Solve Any Two of the following.		12
A)	Design by LSM, a square footing for a column 300mm x 300mm to carry an axial ultimate load of 400 kN. SBC of soil is 400 kN/m ² . Use M-20 grade concrete and Fe-500 grade steel. Draw a neat diagram to show reinforcement details (plan and longitudinal sections)	6	12
B)	A simply supported beam, 3 m in span subjected to u.d.l of 20 kN/m over its entire span. Use M-20 grade concrete, Fe-415 grade steel and 20 mm diameter bars. Adopt size of beam equal to 300 mm x 400 mm (overall). Given $d'=25$ mm. Beam is reinforced with 3 bars of 16 mm at support and mid -section. Design the beam for shear reinforcement. Draw neat diagrams to show reinforcement details in longitudinal section and cross-section. Adopt limit state method .	4	6
C)	Explain the procedure for design of shear for a beam .Draw neat sketch to support your answer.	5	6
Q. 6	Solve Any Two of the following.	5	12
A)	Explain with neat sketches the term: Development length	4	6
B)	Explain the Structural behavior of two-way slab with sketches	3	6
C)	Write step by step procedure for design of a footing.	3	6
	*** End ***		